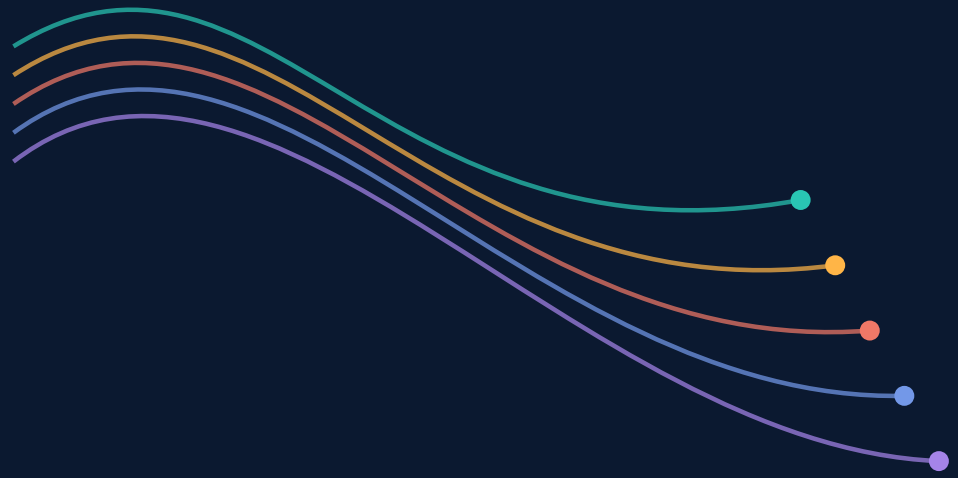
TMS / DATA & ML

# Know before your clients do.



A flight early-warning system for a New York travel agency, built on 435,352 departures from JFK, LGA and EWR. [01]

**2 hours**
notice before scheduled departure [05]

**2 alerts**
late-or-cancelled / cancellation [06]

**1 year**
real flight history · 2023 [01]

THE PROPOSAL

**Pilot two human-reviewed alerts, then decide with evidence—not assumptions.**

THE BUSINESS CASE

# A bad flight is a client problem first

Give the agency time to warn, check connections and prepare alternatives.

## 25.7%

**TRAINING-PERIOD RISK**

Flights late or cancelled, Jan–Sep; 330,883 scheduled flights.

[01]

## 10,738

**CANCELLATIONS**

Every cancellation can force a rebooking decision.

[01]

## 3 airports

**NEW YORK DEPARTURES**

JFK · LaGuardia (LGA) · Newark (EWR).

[01]

THE TWO QUESTIONS

### Will this flight be late or cancelled?

Should the agency prepare a rebooking?

Both scores are intended for the two-hour point. [05]

A person verifies live airline status before acting.

WHAT THE DATA REVEALED

# Risk builds, clusters—and is not just weather

Three findings with a business use. Training-period analysis unless stated. [02, 06]

### THE DAY GETS HARDER

**13.4%** [02]
scheduled 05–08 h



**39.9%** [02]
scheduled 19–22 h



**+26.5 percentage points. [02]**

Tested: delay accumulates along tail turns (p<0.001). [02]

### CANCELLATIONS CLUSTER ON A FEW DAYS

**27 / 273 days** [02] accounted for 57.5% of cancellations. [02]

Peaks in summer convective storms (June 6.1%, July 8.7% vs 0.2% in autumn). [01, 02]

### THE BEST WARNING IS WHAT JUST HAPPENED AT THE AIRPORT

**0.228** recent airport-state importance

**0.023** weather

About 10× stronger in validation permutation importance comparison. [06]

TARGET THE RESPONSE

# Destinations have different reliability profiles

Three clustered groups plus a small-airport label; use the groups as guidance, not certainty.

SHORT-HAUL

**57 airports**

101,287 flights

## 21.2% late

Prepare rebooking options; cancellation rate was 4.3%.

[03]

MAIN LONG-HAUL

**45 airports**

226,896 flights

## 27.8% late

Prioritize afternoon and evening connection checks.

[03]

CHRONIC DELAYS

**7 airports**

2,374 flights

## 53.1% late

Warn early and check onward connections.

[03]

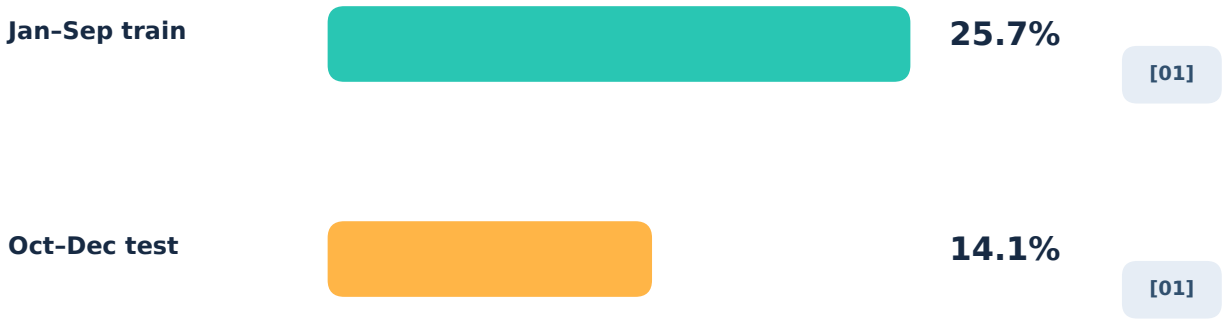
**CAUTION** The structure is modest (silhouette 0.322). Avoid strong claims about small groups. [03]

MEASURE WHAT MATTERS

# The year changes the operating picture

Use a small set of explicit, denominator-aware reliability measures.

LATE OR CANCELLED




Cancellation rate: 3.18% train → 0.21% test. [01]
This drift is why test performance must be shown against its own prevalence.

THE FIVE METRICS

| | | |
|---|---|---|
| M1 | on-time rate | departed flights |
| M2 | mean departure delay | minutes · departed |
| M3 | cancellation rate | all scheduled flights |
| M4 | >60 min late | departed flights |
| M5 | late or cancelled | all scheduled flights |

**Full-year checks: M1 79.03% · M2 13.84 min · M3 2.47% · M4 8.41% · M5 22.91%. [04]**

BUILT TO BE TRUSTED

# Score only with information available at decision time

The prediction point is two hours before scheduled departure. [05]




### SCHEDULE & ROUTE

Calendar · carrier · origin/destination
Scheduled flight volume

### RECENT AIRPORT STATE

Only earlier departures known by T
No earlier cancellation flag

### WEATHER & AIRCRAFT

Strict weather is available by T
Scheduled-hour weather is a forecast stand-in

Leakage guardrails: exclude post-departure outcomes (dep_delay, air_time); plane features follow carrier stats (D10). 4 automated leak tests passed. [05]

AN HONEST TEST

# Tune on validation; keep the test months unseen

Time order mirrors how the system would be built and checked in practice.

### JAN–JUN

Fit candidates
222,402 flights

[06]

### JUL–SEP

Tune and choose
108,481 flights

[06]

### OCT–DEC

Final test · once
104,469 flights

[06]

**Then refit selected settings on Jan–Sep: 330,883 flights. [06]**

WHY NOT ACCURACY?

**“Nobody is late” is 85.9% accurate on test and finds no late flights. [06]**

We compare PR-AUC with the dumb answer on the same period; prevalence is the baseline. [06]

THE RESULT

# Both alerts rank risk above the dumb baseline

Precision–recall AUC on held-out Oct–Dec flights; intervals reflect day-level resampling. [06]

| PREDICTION | DUMB BASELINE | MODEL PR-AUC | 95% INTERVAL | READOUT |
|---|---|---|---|---|
| **Late or cancelled** [06] | **0.141** | **0.339** | 0.301–0.376 | 2.4× prevalence |
| **Cancelled** [06] | **0.0021** | **0.120** | 0.002–0.233 | rare-event caution |

Cancellation scores are useful for prioritization—not certainty. The interval is wide because the test set had only 224 cancellations. [01, 06]

Both models beat the dumb baseline (prevalence) in paired tests ($p < 0.001$). Aircraft features follow D10 carrier-median imputation. [05, 06]

DECISION CARD / ALERT 1

# Warn clients about a possible delay

Late-or-cancelled score · test quarter · human review before messaging. [07]

| .597 | 11.5% | 37.4% | 30.5% |
|---|---|---|---|
| validation cutoff | of test flights flagged | of flags were right | of problem flights caught |
| [07] | [07] | [07] | [07] |

WHAT THE TEST QUARTER LOOKED LIKE

**11,974 flags** **4,481 true positives**

About 130 flagged flights per day; lift 2.7× over a random shortlist. [07]

Capacity view: Top 20% budget flags 227/day (precision 31.2%, recall 44.3%). [07]

ACTION + HUMAN CHECK

## Message · check connections

Verify live airline status first. Mornings are a weak spot: 5.8% recall (evenings 5

Group action: Warn earlier on G2 long-haul evening flights; keep rebooking ready for

DECISION CARD / ALERT 2

# Prepare rebooking for cancellation risk

A prioritization list, not a promise that a flight will be cancelled. [07]

**.137**
validation cutoff
[07]

**484**
test flags
[07]

**19.4%**
flag precision
[07]

**42%**
cancellations caught
[07]

**94 / 224** cancellations identified

5.3 flags per day · lift 91× over random. [07]

**Offer rebooking options; never auto-rebook.**

Capacity view: Top 1% budget flags 11/day (precision 10.2%, recall 47.8%). [07]

**BURST PATTERN & WEAK SPOTS**

**115 cancellations on Dec 18 (121 in ISO Week 51).**

Weather was normal (rain 12% vs 11% median); cause unrecorded. [02, 07]

Weak spot: Newark recall 0% (vs JFK 45%, LGA 54%). Republic flagged 4× base. [06, 07]

LIMITS & MONITORING

# Keep a person in the loop—and watch drift

The test quarter is not a guarantee of future performance.

**SEASONAL DRIFT**

Late-or-cancelled rate fell from 25.7% in train to 14.1% in test. [01]

**EARLY-MORNING GAP**

Alert 1 caught 5.8% of morning problem flights; Newark cancellation recall was 0%. [06, 07]

**RARE EVENT**

115 / 224 test cancellations landed on 18 December; cause is unexplained. [01, 07]

Operational cadence: Weekly drift audit triggers investigation if recall dips below the floor rule. [07]

**A TENTATIVE CHECK RULE**

## Alert 1

Review if weekly recall is below 14% for two weeks running. [07]

## Alert 2

Review 4-week blocks with ≥30 cancellations if recall is below 23%. [07]

Thresholds are tentative; the evidence is limited.

GOVERNANCE & DECISION

# Responsible AI, verified methods—and a careful pilot

Golden Rule #6 compliance, independent audits, and operational next steps. [01–07, README]

## AI ASSISTANCE

Drafted baseline notebook code, helper utilities (utils.py), and visualization layouts; assisted with memory profiling during Task 6 modeling. [README]

## WHAT I INDEPENDENTLY CHECKED

- Recomputed all 435,352 flight counts and 15 metric assertions in code.
- Audited 4 automated leak tests (verified zero future data at T − 2h).
- Executed notebooks 01–07 top-to-bottom with seed 42 (100% reproducible).
- Reconciled every slide figure with the project notebook ledger.

## DATA LIMITATIONS

Single-year data (2023) limits annual seasonality assessment. Strong autumn drift (late rate 25.7% → 14.1%; cancels 3.18% → 0.21%). Weather is observed, not forecast.

## THE DECISION

**Approve a careful pilot.**
**Earn the right to scale.**

| | | |
|---|---|---|
| 01 | REVIEW | A named person verifies live status before client contact. |
| 02 | MONITOR | Weekly drift review; trigger stop rule if recall falls. |
| 03 | IMPROVE | Integrate live forecasts & cancellation timestamps. |

## THE ASK

**Agree the daily alert capacity & assign review owner**

Capacity target: ~130 delay alerts/day (Alert 1) and ~5 rebooking alerts/day (Alert 2). Human ownership is esser

Data source: nycflights23 (CC0). Every metric traceable in notebooks 01–07. All seeds set to 42. [01–07, README]